

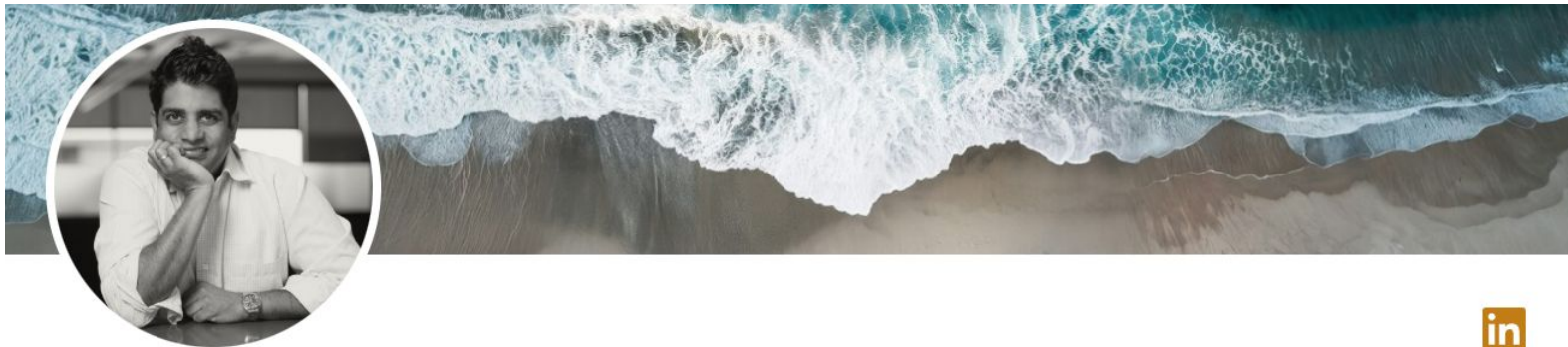
O'REILLY®

Fundamentals of Large Language Models





About me



Jonathan A. Fernandes ✓

AI/ML Engineer Building & Shipping Production-ready GenAI & Large Language Model Solutions Since Before ChatGPT.

United Kingdom · [Contact info](#)



jonfernandes



University of Warwick -
Warwick Business School



Things you need for today

- OpenAI account - <https://platform.openai.com/playground>
- Cohere account - <https://dashboard.cohere.ai/playground>
- Hugging Face account - <https://huggingface.co/>



**What questions about Large Language Models
would you like covered today?**

Please put this in the Q&A

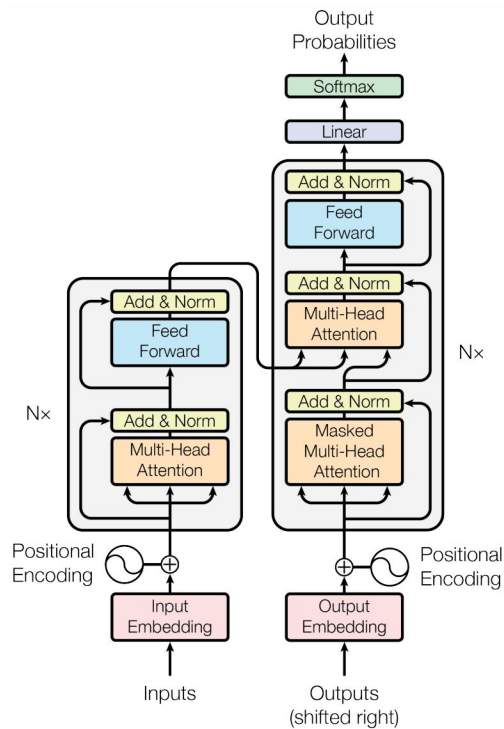


This online training is always being updated.



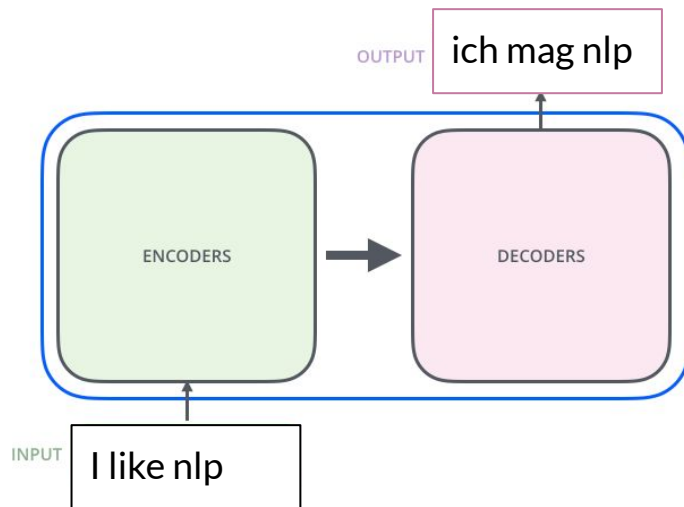
Transformer: Architecture Overview

Transformer architecture

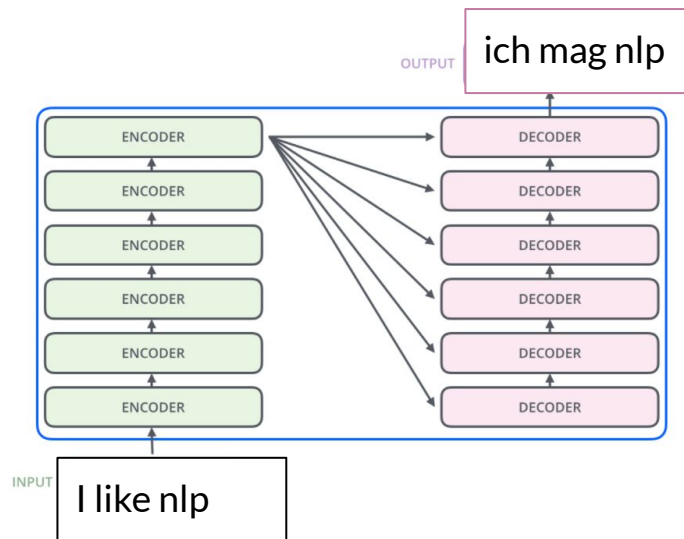




Transformer overview



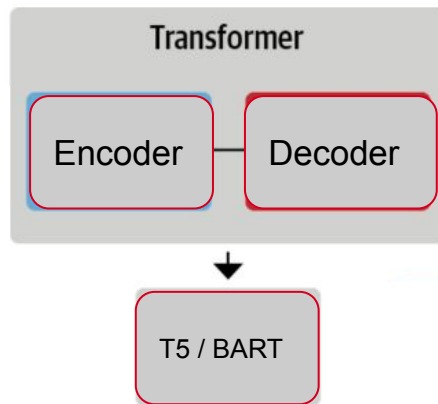
Transformer overview





Encoder-decoder model

- Generative tasks





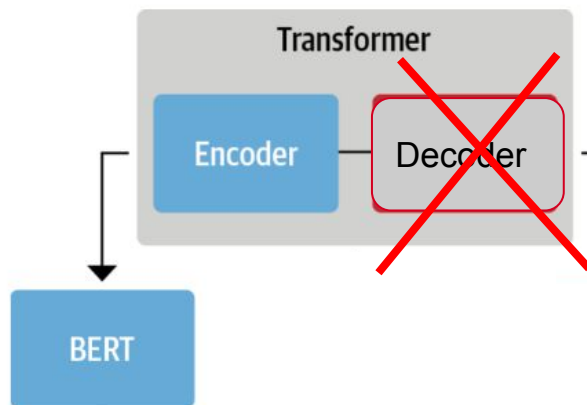
Encoder-only model

Understanding of input

- Sentence classification
- Named Entity Recognition

Family of BERT models:

- BERT, RoBERTa, DistilBERT ...



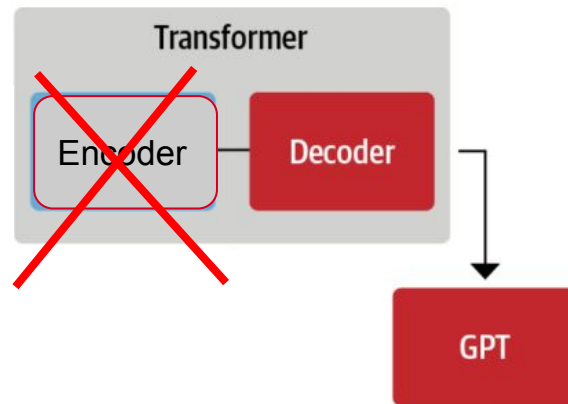


Decoder-only model

- Generative tasks

Examples:

- OpenAI GPT models, Claude, Gemini





Encoder models





Google

curling objective





BERT

Bidirectional Encoder Representations from
Transformers



Where are Transformers used in production?

what's the main objective for curling in the olympics



All



Images



News



Videos



Shopping



More

Tools

About 18,900,000 results (0.65 seconds)

The goal for each team is **to get stones as close to the center of the house as possible and earn points based on the positioning of their stones**. Only one team can score in an end, and points are only awarded if the stones are touching the house. The team with the most points after 10 ends is the winner. 14 Feb 2022

<https://www.sportingnews.com> › olympics › news › curlin... ⋮

[How does curling work? Explaining the rules and scoring for ...](#)



About featured snippets



Feedback



Transformers in production



Can you get medicine for someone pharmacy

BEFORE BERT





Transformers in production

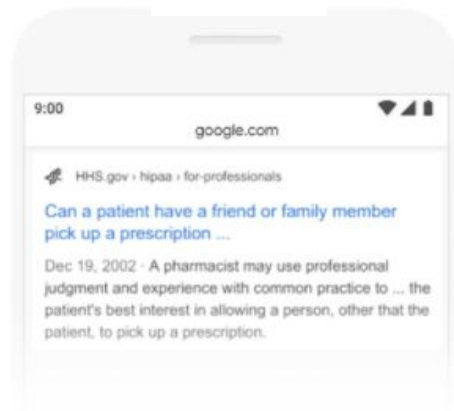


Can you get medicine for someone pharmacy

BEFORE BERT



AFTER BERT





What was BERT trained on?

BERT - Wikipedia and BooksCorpus (11,000 unpublished books)



What tasks was BERT trained?

- Masked Language Model (MLM)
- Next Sentence Prediction (NSP)



The Tokyo Olympic games were <masked> from 2020 to 2021.



Masked Language Modelling (MLM)

The Tokyo Olympic games were <masked> from 2020 to 2021.



Masked Language Modelling (MLM)

The Tokyo Olympic games were postponed from 2020 to 2021.



Next sentence prediction (NSP)

The Tokyo Olympic games were postponed from 2020 to 2021. This is the first instance in the history of the Olympics as previous games had been cancelled but not rescheduled.



Why MLM and NSP?

BERT gets a good understanding of English language.

Pre-training: BERT



	BERT
Year	2018
Number of parameters	109M
Training time	12 days
Infrastructure	8 x V100 GPUs (*)
Size of dataset used for training	16GB
Training tokens (dataset)	250B
Dataset source	Wikipedia
	Book corpus



What are tokens?

1500 words is approximately equivalent to 2400 tokens



What are tokens?

1500 words is approximately equivalent to 2400 tokens

A word is approximately 1.4 tokens



What are tokens?

1500 words is approximately equivalent to 2400 tokens

A word is approximately 1.4 tokens

A novel is 100,000 words, or 140,000 tokens

What are tokens?

BERT was trained on 250B
tokens or:

1.8 million novels





Embeddings



Banana

Basketball

Bicycle

Building

Car

Castle

Cherry

House

Soccer

Strawberry

Tennis

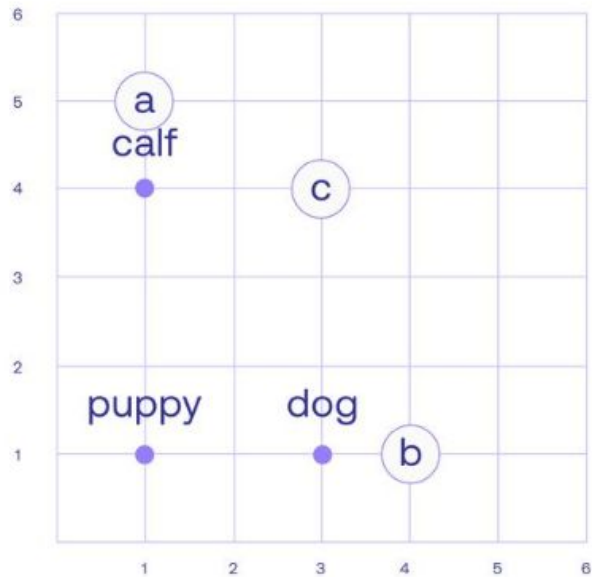
Truck

Embeddings Quiz 1:

Where would you put the word “apple”?



What is c?





Word embeddings

Many more columns

Word	Numbers	
Apple	5	5
Soccer	0	6
House	2	2
Car	6	0



Word	Numbers			
A	-0.82	-0.32	...	-0.23
Aardvark	0.419	1.28	...	-0.06
...			...	
Zygote	-0.74	-1.02	...	1.35

4096



Sentence embeddings with Cohere (demo)

<https://docs.google.com/spreadsheets/d/17AVE0M1mLgOVR1ptDUzP218rVrXbTTzwaQkxDpQlPIQ/edit?usp=sharing>



Similarity between text

- Dot Product
- Cosine Similarity



The more similar two words or sentences are, the larger their Dot Product



Cohere's embeddings have 1024 dimensions



What do each of the dimensions mean?

	Dimension 0 (How citric?)	Dimension 1 (How large?)
Lemons are rich in vitamin C	8	2
Limes are tangy and acidic	9	1
Michael Jordan played for the Chicago Bulls	0	10



What do each of the dimensions mean?

	Dimension 0 (How citric?)	Dimension 1 (How large?)
Lemons are rich in vitamin C	8	2
Limes are tangy and acidic	9	1
Michael Jordan played for the Chicago Bulls	0	10

Dot-product between Lemons and Jordan sentence : $8 \times 0 + 2 \times 10 = 20$



What do each of the dimensions mean?

	Dimension 0 (How citric?)	Dimension 1 (How large?)
Lemons are rich in vitamin C	8	2
Limes are tangy and acidic	9	1
Michael Jordan played for the Chicago Bulls	0	10

Dot-product between Limes and Jordan sentence : $9 \times 0 + 1 \times 10 = 10$



What do each of the dimensions mean?

	Dimension 0 (How citric?)	Dimension 1 (How large?)
Lemons are rich in vitamin C	8	2
Limes are tangy and acidic	9	1
Michael Jordan played for the Chicago Bulls	0	10

Dot-product between Limes and Lemons sentence : $8 \times 9 + 2 \times 1 = 74$



Can we have a similarity score between 0 and 1?

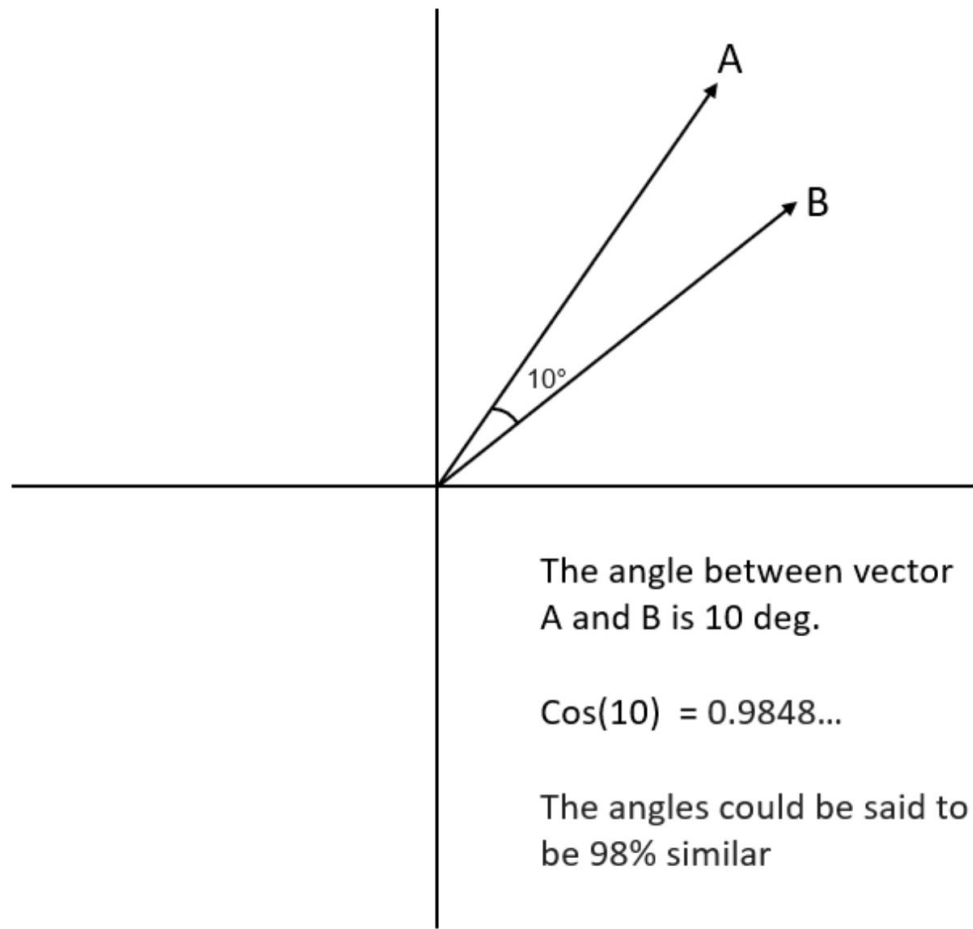


Cosine Similarity:

- 2 sentences that are very dissimilar have a score close to 0.**
- 2 sentences that are similar have a score close to 1.**



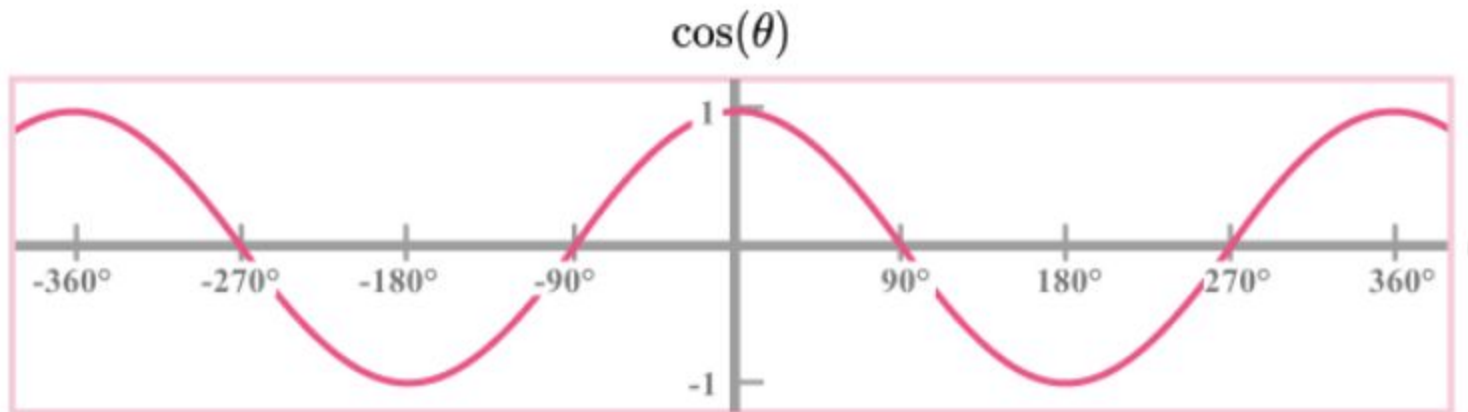
Cosine Similarity





Cosine Similarity:

- 2 sentences that are very dissimilar have a score close to 0.
- 2 sentences that are similar have a score close to 1.





Colab notebook (7 minutes):

<https://colab.research.google.com/drive/1YVy0zrz42z2WexDYUFHMu9XMRIuJgKB5>



Multilingual embedding models



Multilingual demo

https://docs.google.com/spreadsheets/d/11alaXzWwwVk9U8mVjFbGGkoqNzxWBUAGF6tVjB3O_T8/edit?usp=sharing



What are some applications for multilingual embeddings?



What are some applications for multilingual embeddings?

- **Sentiment Analysis:** Analyze customer sentiment in any language.
- **Content Moderation:** Tackle spam and hate-speech in international communities like online gaming.
- **Intent Recognition:** Classify the user's intent based on a set of predefined intents (e.g., booking a flight, ordering food, etc.).



Cross-lingual classification





Anthropic and the Department of War



The Department of War
will only contract with AI
companies who accept
“any lawful use.”

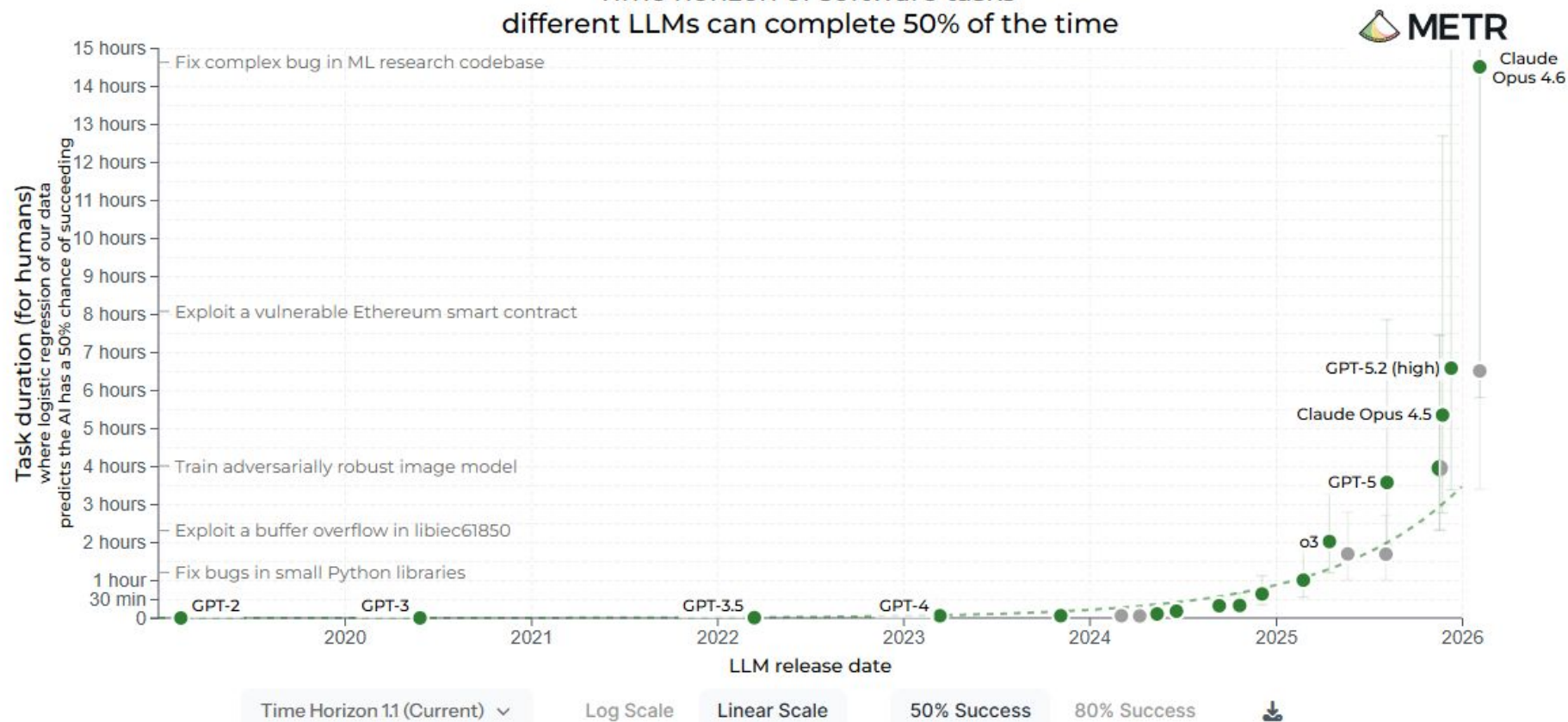
We can't agree to this.

Anthropic supports the lawful use of Claude—
with only two exceptions:

- ① **Mass domestic surveillance**
- ② **Fully autonomous weapons**



Time horizon of software tasks different LLMs can complete 50% of the time



Mass domestic surveillance

The use of AI for mass domestic surveillance presents serious, novel risks to our fundamental liberties.

It is not compatible with democratic values.



Fully autonomous weapons

Current AI systems are not reliable enough to power fully autonomous weapons.

We will not knowingly provide a product that puts America's warfighters and civilians at risk.



The Department of War has threatened to remove us from their systems if we maintain these two exceptions.

Read the full statement from Dario Amodei, our CEO.

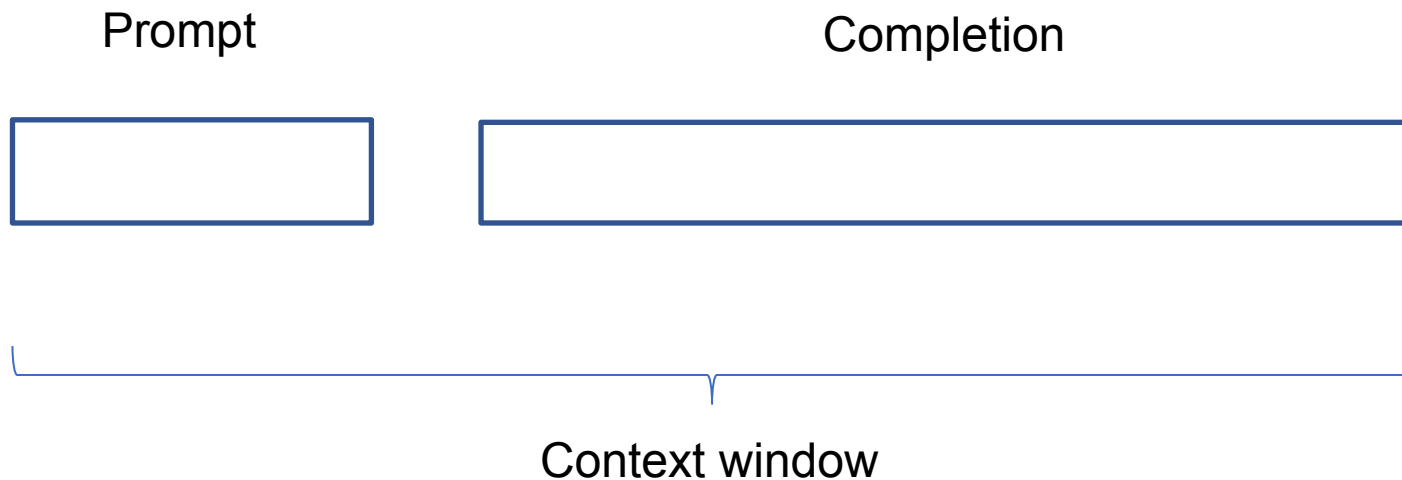
anthropic.com/dow



Context length



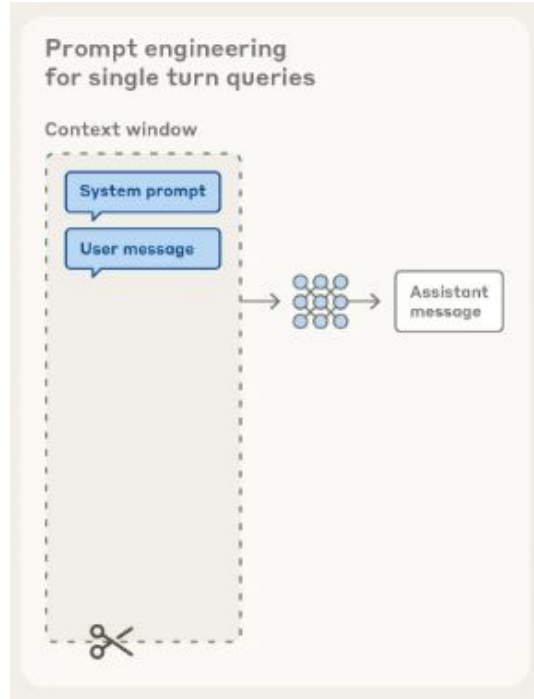
Context window / length





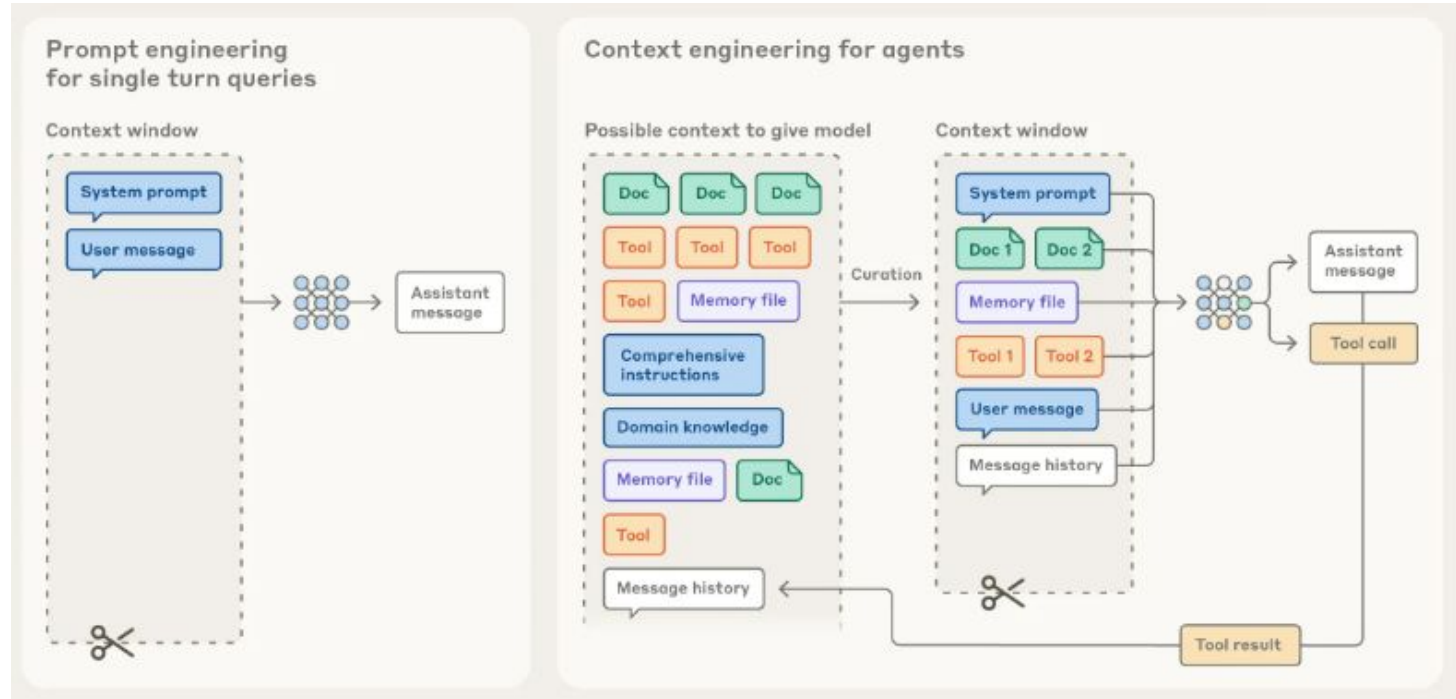
Context Engineering

Difference between Prompt Engineering & Context Engineering





Difference between Prompt Engineering & Context Engineering





Working with context constraints



Context length isn't context depth; LLMs don't actually process the 10,000th token as reliably as the first.



[Haystack]

I've discovered a handy test for figuring out what you're addicted to.

Imagine you were going to spend the weekend at a friend's house on a little island off the coast of Maine. There are no shops on the island and you won't be able to leave while you're there. Also, you've never been to this house before, so you can't assume it will have more than any house might.

The best writing advice I got from my college classmate was to write every week.

What, besides clothes and toiletries, do you make a point of packing? That's what you're addicted to.

For example, if you find yourself packing a bottle of vodka (just in case), you may want to stop and think about that. For me the list is four things: books, earplugs, a notebook, and a pen. There are other things I might bring if I thought of it, like music, or tea, but I can live without them.

I'm not so addicted to caffeine that I wouldn't risk the house not having any tea, just for a weekend.

[Question]

"What was the best writing advice I got from my college classmate?"

[Needle]







Chat Agent

I recently moved to San Francisco, give me
some restaurant recommendations

Here are some good restaurant recommendations
for San Francisco...

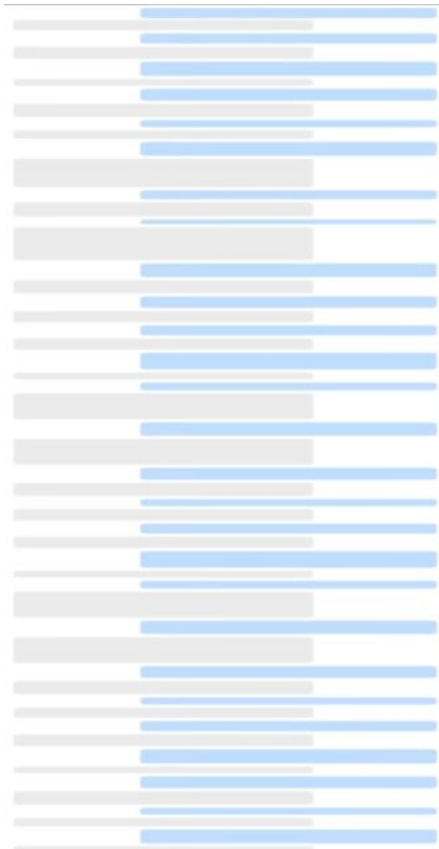


Chat Agent

What are some good outdoor activities to do on this beautiful sunny day?



Naive approach.





~500 Messages



Question



LLM



Answer





~500 Messages



Question



LLM



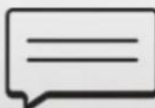
Answer



~2-4 Messages



Question



~300 tokens

LLM



Answer



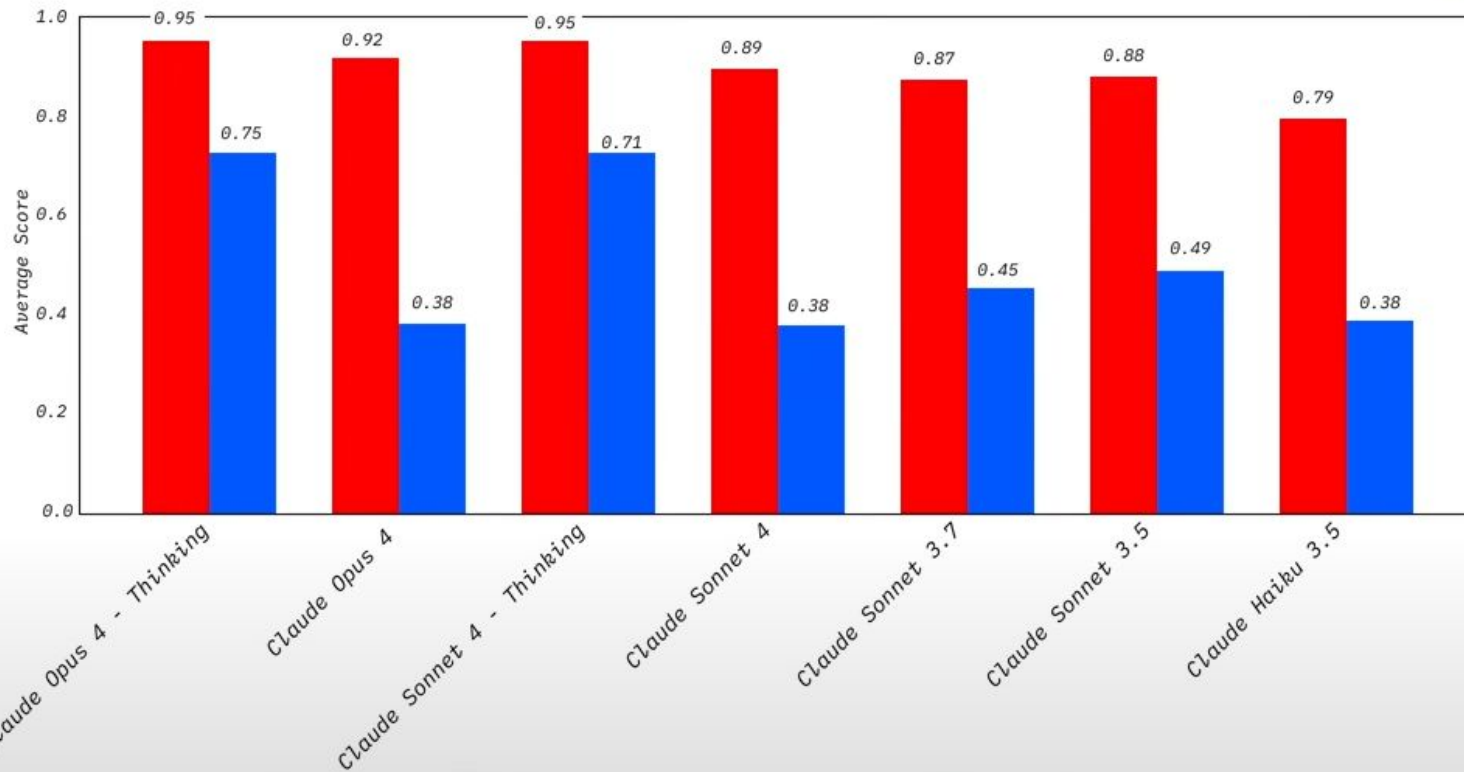


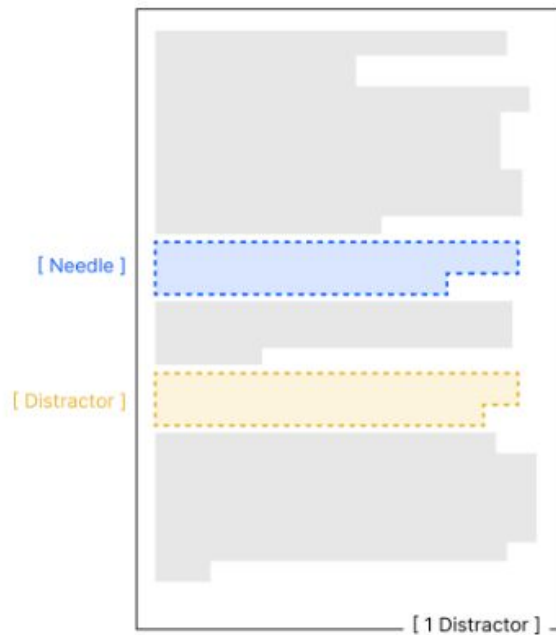
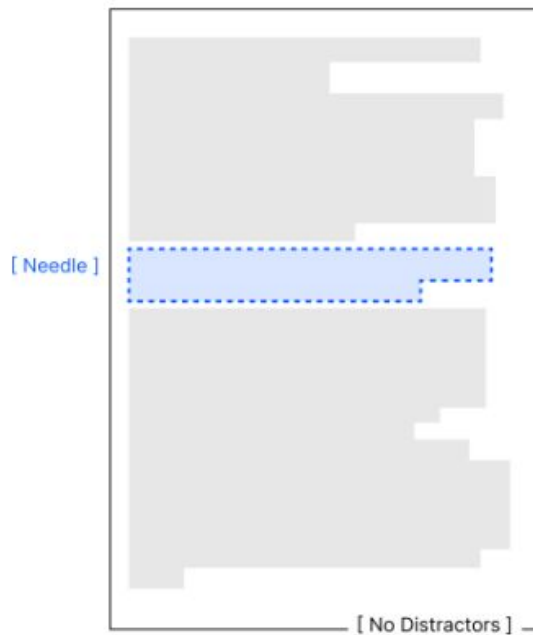
Context length isn't context depth; LLMs don't actually process the 10,000th token as reliably as the first.



LongMemEval Overall Performance - Claude

~2-4 Messages █
~500 Messages █







Distractors



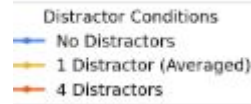
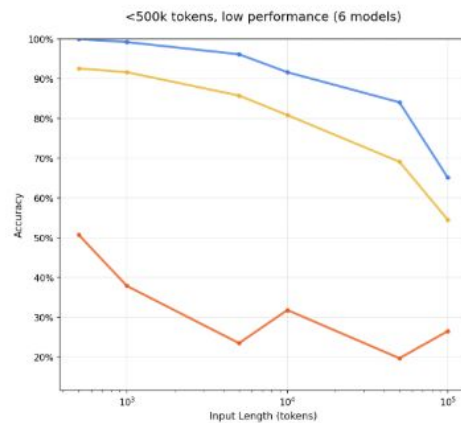
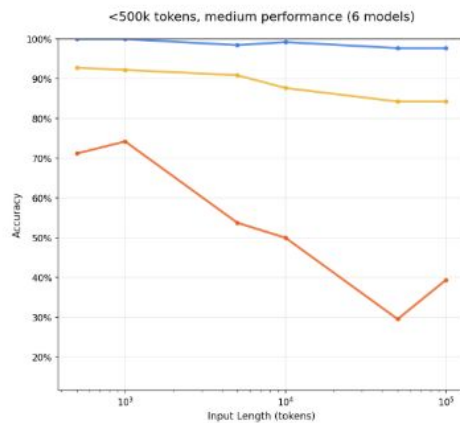
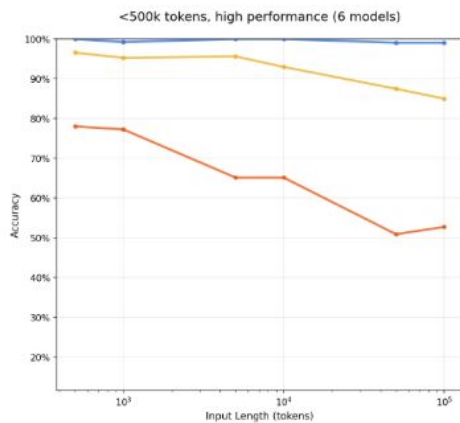
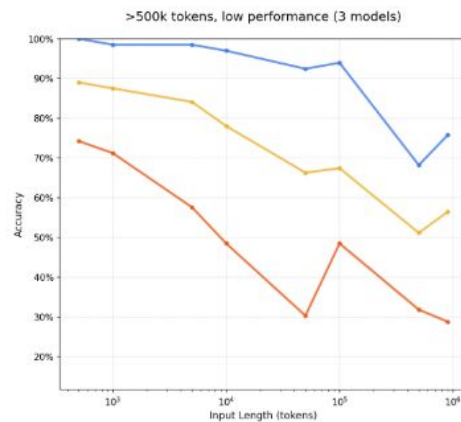
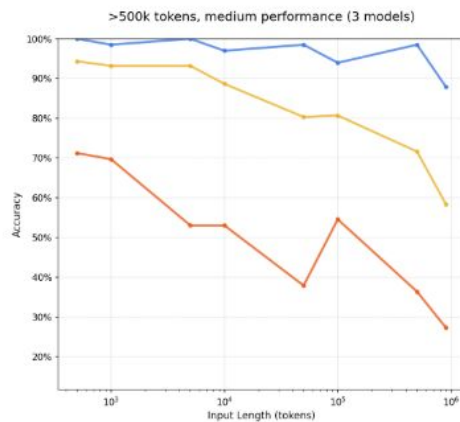
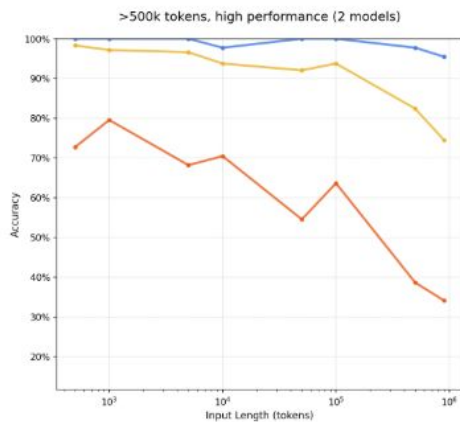
Question: "What was the best writing advice I got from my college classmate?"

Needle: "I think the best writing tip I received from my college classmate was to write every week."

Distractors:

- "The best writing tip I received from my college professor was to write everyday."
- "The worst writing advice I got from my college classmate was to write each essay in five different styles."
- "The best writing advice I got from my classmate was to write each essay in three different styles, this was back in high school."

Distractors





MCP



Why MCP?

- Creates a universal language for AI Agents
- Allows networks of tools and models to work together
- Standardizes communication
- Adoption



Working with LLMs



Task performance is inconsistent

- Excellent at creative writing and drafting emails
- Inadequate for high precision tasks



What is
 1234567×1234576



Working with LLMs

- Demo
 - GPT-4-Turbo
- After Demo
 - Why don't we hand off calculations to calculator?



Using tools with LLMs



Using tools with LLMs

- LLMs need to determine ***what*** needs to be done, not ***how*** to actually do it

Demo:

- GPT-4-Turbo (without Code Interpreter)
- gpt-4o + Code Interpreter
- It's a sunny day



Working with APIs



Working with APIs

Demo:

- GPT-4-Turbo currency conversion
- API - Application Programming Interface
- APIs provide a standard way to access services
- APIs are everywhere.
- Need to give tools access to APIs



Working with APIs

Demo:

- GPT-4-Turbo (function) - Capture the Base & Target currency using 3-letter code and Amount
- Free text - Whatever they use in the UK
- <https://www.exchangerate-api.com/> (jafdx)
 - Pair conversion
 -



Working with APIs

- Tooling is not scaleable
(multiplier, currency conversion)
- Require significant maintenance (v6)
- Security and privacy considerations
(Bank accounts, health records)



Working with APIs

- Tooling is not scaleable
(multiplier, currency conversion)
- Require significant maintenance (v6)
- Security and privacy considerations
(Bank accounts, health records)

... and these are the issues that MCP addresses.



Using LLMs with MCP



Using LLMs with MCP

Protocol	How does It Work?	Use Cases
HTTP	TCP Connection HTTP REQ HTTP RESP	 Web Browsing
HTTP/3 (QUIC)	UDP Connection 1 2 3 4	 IoT Virtual Reality
HTTPS	TCP Connection public key session key encrypted data	 Web Browsing
WebSocket	HTTP Upgrade Full Duplex	 Live Chat Real-Time Data Transmission
TCP	SYN SYN + ACK ACK	 Web Browsing Email Protocols
UDP	REQUEST RESPONSE	 Video Conferencing
SMTP	sender SMTP Server receiver	 Sending/Receiving Emails
FTP	Control Channel Data Channel	 Upload/Download Files



Using LLMs with MCP

<https://modelcontextprotocol.io>



Host: The user-facing side

The Host's job is to:

- Handle the user interface and permissions
- Kick off connections to MCP Servers via Clients
- Manage the back-and-forth between the user, the LLM, and any connected tools
- Present the final output clearly to the user



Client: The Connector

The Client is a behind-the-scenes component inside the Host. It:

- Maintains a one-to-one connection with a Server
- Speaks the MCP protocol
- Translates requests from the Host into messages the Server understands



Server: The Tool Provider

The Server is an external system that offers services or data to the AI model. It:

- Gives the model access to tools or databases
- Wraps existing functionality in a lightweight, AI-friendly interface
- Can be hosted locally or remotely
- Makes its features discoverable and usable through MCP



Using LLMs with MCP

At its core, MCP is designed to embed your workflow into AI applications, giving essential context to any system using a large language model. This context could come in the form of tools or even just raw data.



Using LLMs with MCP



What do Large Language Models interact with?

They don't interact directly with APIs. They interact with prompts and tools and whatever you're giving the model to ingest.



Coding lab



Creating an app

- Required subscription
- Simple app (so creating it won't take long).
 - Create your own app. You don't have to use mine.
- Github account (and Github Pages)
- Basic knowledge of git



App



INITIAL

- Demo working example at /life
- /demo/[app.py](#) uses the Python flask framework to create an interactive webpage
- GitHub Pages doesn't support flask
- Using /demo/[app.py](#) file create an equivalent css/html page

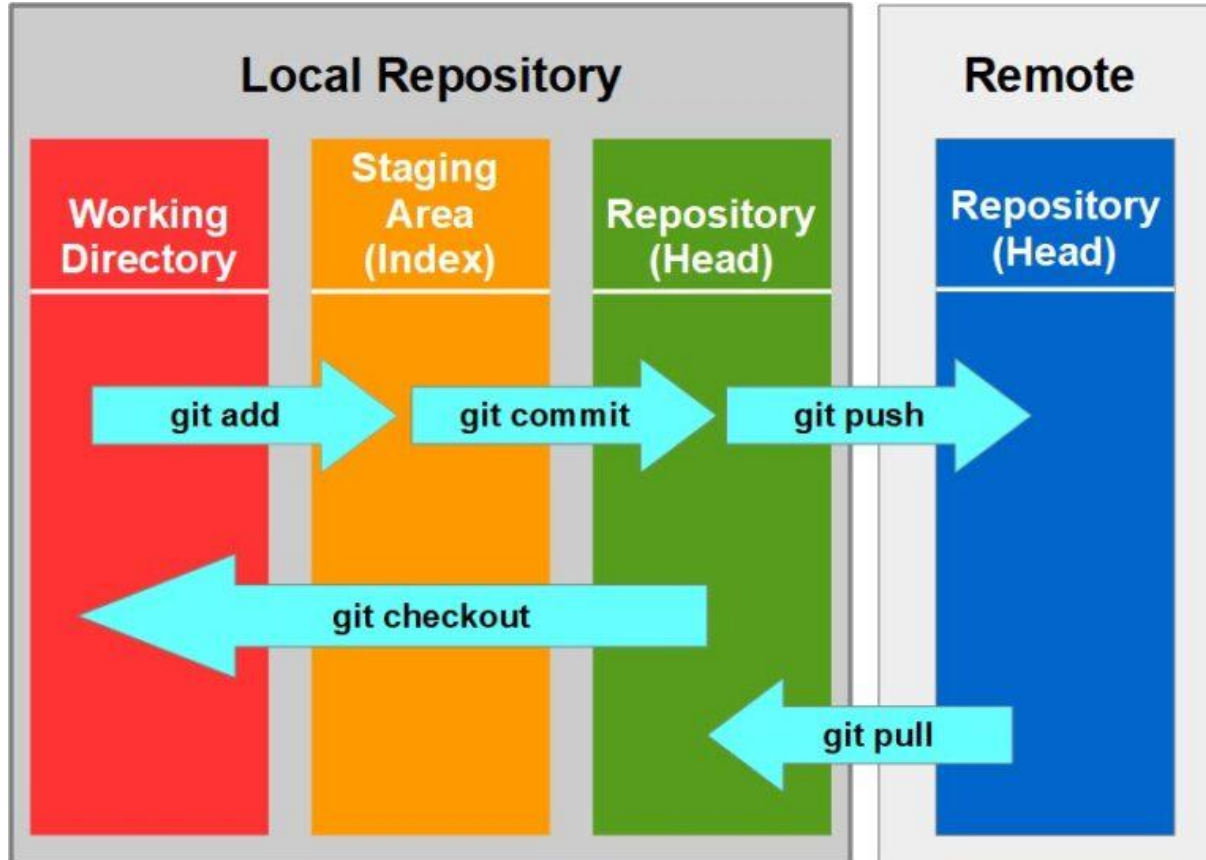
OPTIONAL

- Display images a screenful at a time
- Support dark mode

What codex needs to know

The file demo/app.py uses flask. Change it so that I have the same look and feel using css and html. I want the final files to run on jonfernandes.github.io/demo. Push the changes to Github

Git basics



The background is a gradient from red-orange on the left to yellow on the right. Three large, semi-transparent circles are overlaid: a large red-orange one on the left, a medium red-orange one in the center, and a smaller yellow one on the right. The text "O'REILLY" is centered in white, bold, sans-serif font, with a registered trademark symbol (®) at the end.

O'REILLY®

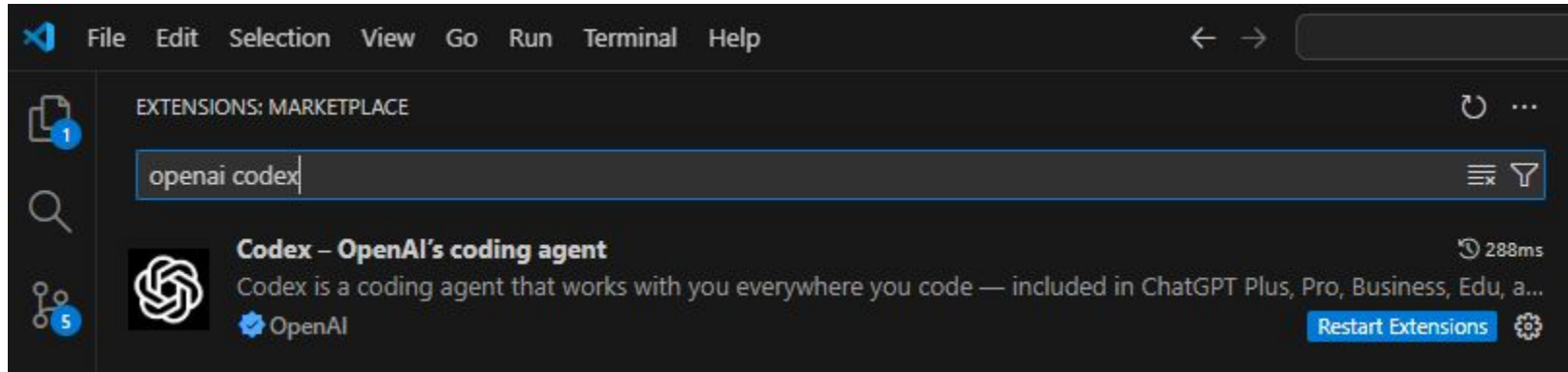


Codex app



Code editor

Using code editor





CLI – Command Line Interface

Codex CLI



1 Install

Install the Codex CLI with npm.

```
npm i -g @openai/code:
```



2 Run

Run Codex in a terminal. It can inspect your repository, edit files, and run commands.

```
codex
```



The first time you run Codex, you'll be prompted to sign in. Authenticate with your ChatGPT account or an API key.

See the [pricing page](#) if you're not sure which plans include Codex access.

3 Upgrade

New versions of the Codex CLI are released regularly. See the [changelog](#) for release notes. To upgrade with npm, run:

```
npm i -g @openai/code:
```



The Codex CLI is available on macOS and Linux. Windows support is experimental. For the best Windows experience, use Codex in a WSL workspace and follow our [Windows setup guide](#).



Working with Codex CLI

Run Codex interactively

Run `codex` to start an interactive terminal UI (TUI) session.

Control model and reasoning

Use `/model` to switch between GPT-5.3-Codex and other available models, or adjust reasoning levels.

Image inputs

Attach screenshots or design specs so Codex reads them alongside your prompt.

Run local code review

Get your code reviewed by a separate Codex agent before you commit or push your changes.

Web search

Use Codex to search the web and get up-to-date information for your task.

Codex Cloud tasks

Launch a Codex Cloud task, choose environments, and apply the resulting diffs without leaving your terminal.

Scripting Codex

Automate repeatable workflows by scripting Codex with the `exec` command.

Model Context Protocol

Give Codex access to additional third-party tools and context with Model Context Protocol (MCP).

Approval modes

Choose the approval mode that matches your comfort level before Codex edits or runs commands.



Working with Codex CLI

```
/
/model          choose what model and reasoning effort to use
/permissions    choose what Codex is allowed to do
/experimental    toggle experimental features
/skills         use skills to improve how Codex performs specific tasks
/review         review my current changes and find issues
/rename         rename the current thread
/new            start a new chat during a conversation
/resume         resume a saved chat
```



Working with Codex CLI

```
/
/init      create an AGENTS.md file with instructions for Codex
/compact  summarize conversation to prevent hitting the context limit
/plan      switch to Plan mode
/collab    change collaboration mode (experimental)
/agent     switch the active agent thread
/diff      show git diff (including untracked files)
/mention   mention a file
/status    show current session configuration and token usage
```



AGENTS.md using /init

AGENTS.md

A simple, open format for guiding coding agents, used by over [60k open-source projects](#).

Think of AGENTS.md as a **README for agents**: a dedicated, predictable place to provide the context and instructions to help AI coding agents work on your project.

[Explore Examples](#)[View on GitHub](#)

```
# AGENTS.md

## Setup commands
- Install deps: `pnpm install`
- Start dev server: `pnpm dev`
- Run tests: `pnpm test`

## Code style
- TypeScript strict mode
- Single quotes, no semicolons
- Use functional patterns where possible
```

<https://agents.md>

Why AGENTS.md?

README.md files are for humans: quick starts, project descriptions, and contribution guidelines.

AGENTS.md complements this by containing the extra, sometimes detailed context coding agents need: build steps, tests, and conventions that might clutter a README or aren't relevant to human contributors.

We intentionally kept it separate to:

- 📖 Give agents a clear, predictable place for instructions.
- 👤 Keep READMEs concise and focused on human contributors.
- 🔗 Provide precise, agent-focused guidance that complements existing README and docs.

Rather than introducing another proprietary file, we chose a name and format that could work for anyone. If you're building or using coding agents and find this helpful, feel free to adopt it.



AGENTS.md

Repository Guidelines

Project Structure & Module Organization

This project is a small static web app under `noah3/`.

- `index.html`: page structure and UI screens.
- `styles.css`: layout, theme, responsive behavior.
- `app.js`: quiz logic, scoring, and learning-mode flow.
- `JF-color-icon.jpg`, `JF-bw-square.png`: image assets used by the UI.

Keep new code in these files unless a feature clearly needs a new module (for example, `question-bank.js`). Place additional images in `noah3/` and reference them with live paths.

Build, Test, and Development Commands

No build pipeline is required; this is plain HTML/CSS/JS.

- `python3 -m http.server 8000` (from `noah3/`): run locally at `http://localhost:8000`.
- `node --check app.js`: quick JavaScript syntax validation.
- `git -C .. status --short noah3`: inspect changes limited to this app folder.

Coding Style & Naming Conventions

- Use 2-space indentation in HTML/CSS/JS to match existing files.
- Prefer descriptive camelCase for JavaScript variables/functions (for example, `startLearningMode`).
- Use kebab-case for CSS classes and IDs only when already established (for example, `screen-summary`, `quit-btn`).
- Keep UI text child-friendly and concise.
- Avoid adding dependencies unless there is a clear need.

Testing Guidelines

There is no automated test suite yet.

- Run `node --check app.js` before each commit.
- Manually verify key flows in browser:
 1. 10-question assessment runs end-to-end.
 2. Weak-topic summary appears.
 3. Learning mode caps at 10 questions.
 4. `Quit` returns to home from any active screen.

Commit & Pull Request Guidelines

Recent history uses short, imperative commit messages (for example, `Add quit button across quiz flow`). Follow that pattern.

For pull requests:

- Summarize user-visible changes.
- List files touched (for example, `noah3/app.js`, `noah3/styles.css`).
- Include screenshots/GIFs for UI updates.
- Note manual test steps performed.